Ziqi Liu

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Education

- **The Hong Kong University of Science and Technology** *Sep 2025 – Expected Jun 2027*
  Master of Philosophy, Integrated Systems and Design Hong Kong
  - **Advisor:** Qijia Shao
  - **Research Focus: Mobile & Ubiquitous Computing, HCI.**
  - *I focus on wearable and mobile sensing technologies, adaptive interactive systems, and mobile health, aiming to design AI-assisted systems that implicitly sense users' behavioral and physiological states and provide timely, personalized, context-aware interventions.*

- **Tsinghua University** *Sep 2021 – Jun 2025*
  Bachelor of Engineering, Intelligent Engineering and Creative Design Beijing, China
  - **Advisors:** Yuntao Wang, Haipeng Mi
  - GPA: 3.70/4.0
  - *An interdisciplinary undergraduate program that combines Electrical Engineering and Computer Science (EECS, main part), Mechanical Engineering (ME), and Industrial Design (ID).*

Publications

[1] **Ziqi Liu**, Ziyi Xu, Jinhe Wen, Xiangjie Tang, and Qijia Shao. 2026. **PreTap: Implicit Reachability Analysis and Tap Prediction for One-Handed Mobile Interaction.** *In Proceedings of the ACM on Interactive, Mobile, Wearable and Ubiquitous Technologies* **(Accepted by IMWUT '26)**.

[2] Xiangjie Tang, **Ziqi Liu**, Minhao Cui, and Qijia Shao. 2026. **Turning Phone-Surface Contacts into Probabilistic Landmarks for Indoor Localization.** Submitted to *ACM Conference on Embedded Networked Sensor Systems* **(Submitted to SenSys '27)**.

[3] Yibo Wang, **Ziqi Liu**, Jiao Xue, and Qi Lu. 2025. **AroMR: Decentralizing Olfactory Displays into the Environment for Olfactory-Augmented Experiences in Mixed Reality.** In *Extended Abstracts of the CHI Conference on Human Factors in Computing Systems* **(CHI EA '25)**. [Paper]

[4] Chang Liu, Xiangyang Wang, Chun Yu, Yingtian Shi, Chongyang Wang, **Ziqi Liu**, Chen Liang, and Yuanchun Shi. 2025. **Enhancing Smartphone Eye Tracking with Cursor-Based Interactive Implicit Calibration.** In *Proceedings of the 2025 CHI Conference on Human Factors in Computing Systems* **(CHI '25)**. [Paper]

Research Experience

- **UbiquitousX Lab, The Hong Kong University of Science and Technology** *09/2025 – Now*
  M.Phil Student / Advisor: Prof. Qijia Shao Hong Kong
  - Currently investigating how physical pressure, body posture, and related contextual factors affect wrist-worn PPG signals, aiming to support more robust signal recovery, physiological state sensing, and downstream mobile health applications.
  - Led projects related to user-state sensing and interaction-assistance on smartphones, including *PreTap* (**IMWUT '26**) and *STEP* (submitted to **SenSys '27**), developing on-device AI systems that leverage implicit motion sensing to infer user-specific states and provide personalized interaction support.

- **Pervasive HCI Group, Tsinghua University** *05/2024 – 06/2025*
  Undergraduate Research Assistant / Advisor: Prof. Chun Yu, Prof. Yuntao Wang Beijing, China
  - Conducted algorithm design for *COMETIC* (**CHI '25**), which introduced a cursor-based implicit calibration method for smartphone eye tracking that leverages correlations between cursor operations and gaze patterns to continuously fine-tune tracking accuracy during natural interaction.
  - Designed a pen-based sensing system that integrates EDA, PPG, and IMU sensors to enable unobtrusive physiological monitoring and cognitive load assessment during natural handwriting interaction.

- **Future Lab, Tsinghua University** *03/2024 – 01/2025*
  Undergraduate Research Assistant / Advisor: Prof. Haipeng Mi, Prof. Qi Lu Beijing, China
  - Co-led *AroMR* (**CHI EA '25**), which introduces a field-centric olfactory rendering strategy by decentralizing displays into the environment to create immersive, context-aware scent experiences in Mixed Reality.
  - Conducted interaction and system design to explore the redesign possibilities of traditional embroidery using electroluminescent threads, proposing a framework for sustainable cultural preservation through smart materials.

Industry Experience

- **Shokz** *06/2025 – 08/2025*
  Research Intern Shenzhen, China
  - **Project:** Design and Deployment of Clamping Force Test Sensors for Head-Mounted Headphones
  - Developed clamping force measurement systems and wearable test prototypes for two bone conduction headphone products, and conducted user and wearing tests to support pre-research application scenarios.
- **XIAO MI** *01/2025 – 03/2025*
  Product Manager Intern, Corporate Group Technology Committee – Xiao Ai Interconnection Group Beijing, China
  - Designed and implemented voice interaction response strategies for scenarios with coexistence of multiple devices such as smartphones, smart home devices, and wearable devices.
- **Mercedes Benz (Beijing) & Tsinghua University** *08/2024 – 11/2024*
  Research Intern Beijing, China
  - **Project:** Towards Sustainable Car Interior Design with Smart Interactive Material
  - Designed and fabricated the high-fidelity demo for interior design with interactive materials, responsible for lighting effects design and circuit implementation.

Teaching Experience

- **ISDN5230 – Artificial Intelligence of Things for Healthcare** *Spring 2026*
  Teaching Assistant / Instructor: Qijia Shao Hong Kong

Academic Service

- **ACM IMWUT** *10/2026*
  Student Volunteer Shanghai, China
- **ACM MobiCom** *11/2025*
  Student Volunteer Hong Kong

Advising and Collaborating

- **Yin Cao**, Undergraduate Student in ECE, University of Florida *Present*
- **Xiangjie Tang**, Undergraduate Student in CS, Southeast University; now CS Ph.D., SNU *2025 – 2026*
- **Ziyi Xu**, Undergraduate Student in ISD, HKUST; now Ph.D., HKUST *2025 – 2026*

Skills

- **Programming Languages:** Python, C/C++, HTML, shell
- **Embedded Systems & Hardware Design:** Raspberry Pi, ESP32, Arduino, Verilog, QuartusII, Multisim
- **Modeling & Graphic Design:** AutoCAD, Solidworks, Fusion 360, Figma, Adobe suite, Unity, Blender
- **Skill Set:** Machine Learning (PyTorch), User Interface Design, Digital Fabrication, Rapid Prototyping